Mohit Appari

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Summary

AI/ML Engineer with 5 years of experience in Python building scalable data infrastructure, ML models, and forecasting systems over millions of records and $200M+ in payments. Skilled in time-series modeling, statistical research, and hypothesis testing that turn large-scale data into high-stakes financial decisions. Driven by quantitative finance, economics, and mathematics.

Experience

Data Scientist Jun 2025 – Present

Florida Agency for Persons with Disabilities

- Built and optimized **15+** automated Python/SPSS pipelines across Medicaid claims, FMMIS, iConnect, and Azure, tuning SQL over millions of records to cut processing time by **83%**.
- Forecasted expenditure across **$200M+** in payments using ARIMA, regression, and XGBoost/LightGBM, within **5–10%** of actuals, to guide budgeting and rate decisions.
- Initiated and led Power BI dashboards on service utilization, regional trends, and spending, used by executive leadership for programs serving **60K+** individuals.
- Detected billing outliers and ran hypothesis tests and A/B tests in Python and SPSS to answer stakeholder questions about consumers, providers, and payments, providing evidence-based insights for program and payment decisions.

AI/ML Researcher Jun 2024 – Sep 2024

Florida State University

- Fine-tuned NequIP, an equivariant GNN, on **11K+** material structures with a custom weighted loss, cutting validation error by **15%**.
- Built reproducible training pipelines with Python, Docker, MLflow, and HPC, enabling tracked, repeatable experiments.
- Selected the best model architecture through controlled experiments, hypothesis testing, and out-of-sample validation.

Software Developer Jan 2022 – Aug 2023

S&P Global

- Led the migration of legacy .NET document-processing workflows to a scalable open-source Python extraction platform, cutting licensing costs by **80%**.
- Mentored a **5**-person team and helped architect and roll out systems processing **10K+** financial, energy, and automotive documents per day at **95%** extraction accuracy.
- Built OCR and CNN classification workflows with Tesseract and PyTorch, converting unstructured financial filings and earnings data into structured datasets for downstream analytics.
- Engineered fault-tolerant AWS/Docker pipelines with event-driven ingestion, retries, validation, and GitHub Actions CI/CD at **80%** test coverage, reducing manual extraction by **90%+**.

Software Developer Intern May 2021 – Aug 2021

LG Electronics

- Built real-time drowsiness detection and logistics tracking with Python, OpenCV, and YOLO, hitting **90%** accuracy, cutting sorting time by **25%**, and placing **3rd** in LG's internal Ideathon.

Software Developer Intern Jan 2021 – Mar 2021

ITC Infotech

- Developed Flask REST APIs with MySQL for master-data management, role-based access, and automated product tracking, reducing manual workload by **30%**.

Projects

BayWheels Trip Analytics Platform | *Python*, *Kafka*, *dbt*, *Terraform*, *GCP (BigQuery, GCS)*, *Docker*

- Built an end-to-end batch pipeline (Terraform, Prefect, dlt, DuckDB, dbt) loading **400K+** monthly Lyft ride records into partitioned BigQuery for Looker analytics.
- Added a real-time streaming layer with Kafka, InfluxDB, and Grafana to monitor live ride events for demand and mobility analysis.

Technical Skills

AI/ML: LLMs, RAG, MCP, AI Agents, LangChain, LangGraph, LlamaIndex, Pinecone, Ollama, scikit-learn, W&B
Quant Finance: Derivatives, Options Pricing, Greeks, Volatility, Financial Modeling, Equity Analysis, Alpha Research
Data & Infra: Snowflake, Spark, Kafka, Airflow, dbt, Hadoop, Delta Lake, DuckDB, Oracle, MongoDB, Redis, Cassandra, Neo4j, GCP, Kubernetes, Terraform, Linux, Bash

Education

Florida State University Tallahassee, FL

*Master of Science in Data Science; GPA: **4.0/4.0*** *Aug 2023 – May 2025*

- **Relevant coursework:** Probability, Statistics, Linear Algebra, Stochastic Processes, Machine Learning, Business Analytics, Financial Management